

Python midterm 2026-2027 - A

April 11, 2026

Duration: 60 minutes

Name: _____ **Neptun:** _____

Theory: 15 questions $\times$ 2 pts = 30 pts. **Practical:** 10 pts

1. What is the expected result of executing the following code?

```
class Counter:
    def __init__(self, start):
        self.value = start
        print(f"Init:{self.value}")

    def step(self, amount):
        try:
            if amount < 0:
                raise ValueError("Negative step")
            self.value += amount
            print(f"Try:{self.value}")
            return self.value
        except ValueError:
            self.value = 0
            print(f"Except:{self.value}")
        finally:
            self.value += 1
            print(f"Finally:{self.value}")
```

```
c = Counter(5)
print("A", c.step(2))
print("B", c.value)
```

Output: _____

2. What is the expected result of executing the following code?

```
def collect_digits(n, acc=[]):
    if n == 0:
        acc.append(0)
        print("Base:", acc)
        return 0
```

```

    digit = n % 10
    result = digit + collect_digits(n // 10, acc)

    acc.append(result)
    print("Step:", acc)
    return result

print("A", collect_digits(23))

```

Output: _____

3. What is the expected result of executing the following code?

```

data = {
    True: "A",
    1: "B",
    False: "D",
    0: "E"
}

print("Dict:", data)

print("X", data[True])
print("Y", data[1])

```

Output: _____

4. What is the expected result of executing the following code?

```

def check(x):
    return x

values = [1, 0, "0", []]

for v in values:
    result = isinstance(v, int) and check(v)
    print(v, "->", result)

```

Output: _____

5. What is the expected result of executing the following code?

```

def process(a, b):
    a.append(100)
    b = b + [200]
    print("Inside:", a, b)

x = [1, 2]
y = x
z = x[:]

```

```
process(y, z)

print("x:", x)
print("y:", y)
print("z:", z)
```

Output: _____

6. What is the expected result of executing the following code?

```
def modify(x):
    x += [11]
    return x

lst = [1, 2, 3]

for item in lst:
    if item % 2 == 0:
        lst = modify(lst)
        break
    else:
        modify(lst)

print("Final:", lst)
```

Output: _____

7. What is the expected result of executing the following code?

```
any_data = ["hello", "", 0, -1]

for el in any_data:
    if el:
        print(el)
```

Output: _____

8. What is the expected result of executing the following code?

```
my_list = [1, 2, 3, 4, 5]

def fun(lst):
    del lst[lst[1]]
    return lst

print(fun(my_list))
```

9. What is the expected result of executing the following code?

```
def process(data):
    return data[1::]

any_data = [{"a":1, "b":2 }, "", -2, 0]
```

```
while any_data:
    any_data = process(any_data)
    print(any_data)
```

Output: _____

10. What is the expected result of executing the following code?

```
def f():
    return [1, 2], [3, 4]
```

```
z = f()
x, y = z
x.append(10)
```

```
print(x, y)
print(type(z))
```

Output: _____

11. What is the output of the following loop?

```
for el in range(4, -1, -1):
    if el % 2 == 0:
        el += 1
    print(el)
```

Output: _____

12. Predict the output of the following code. For each printed line, explain why the type is what it is.

```
f = open("data.txt", "r")
result = f.read()
f.close()
```

```
def func1():
    return "Hi"
```

```
def func2():
    return 10
```

```
items = [result, func1(), func2(), f, func1]
```

```
for el in items:
    print(el, "->", type(el))
```

Output: _____

13. What is the expected result of executing the following code?

```

text = "Python Prog"

result = ""

for i in range(len(text)):
    if text[i].lower() in "aeiou":
        result += text[i]
    elif i % 2 == 0:
        result += text[i].upper()
    else:
        result += text[i].lower()

print(result[::-1])

```

Output: _____

14. What is the expected result of executing the following code?

```

a = [1, 2, 3]
b = a
c = a[:]

print(a is b)
print(a is c)
print(a == b)
print(a == c)

```

Output: _____

15. What is the expected result of executing the following code?

```

class Box:
    def __init__(self, value):
        self.value = value

a = Box(10)
b = a
c = Box(10)

print(id(a) == id(b))
print(id(a) == id(c))

b.value = 20

print(id(a) == id(b))
print(id(a) == id(c))
print(a.value)
print(b.value)

```

```
print(c.value)
```

Output: _____

Practical task(10 points):

Create a class **BookPile** that uses a list internally and supports the following operations:

- **add_book(title)** → adds a book to the top of the pile
- **remove_book()** → removes and returns the top book
- **top_book()** → returns the top book without removing it
- **is_empty()** → returns True if the pile is empty
- **show_books()** → prints all books in the pile

Additional requirements:

- If **remove_book()** or **top_book()** is called when the pile is empty, print: **“No books in the pile”** and return None.

Example usage:

```
pile = BookPile()

pile.add_book("Python Basics")
pile.add_book("Data Science")

pile.show_books()
print("Top book:", pile.top_book())

pile.remove_book()
pile.remove_book()
pile.remove_book()
```